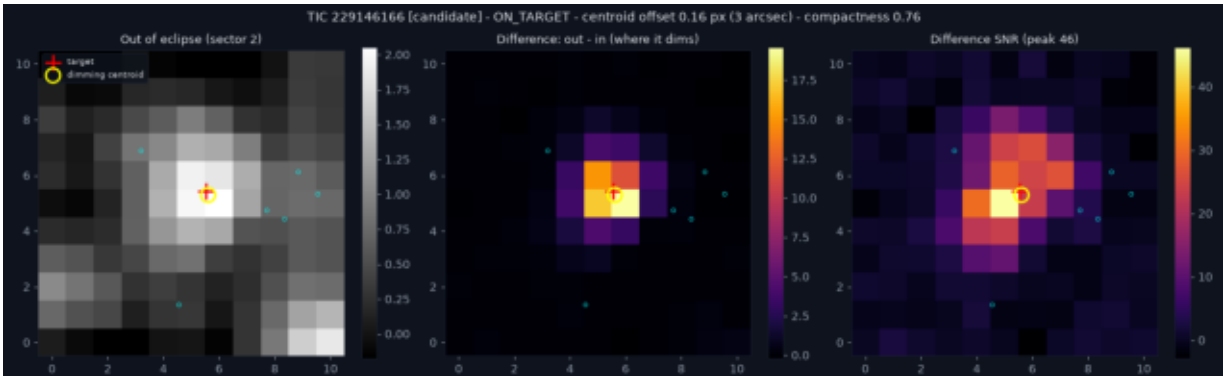
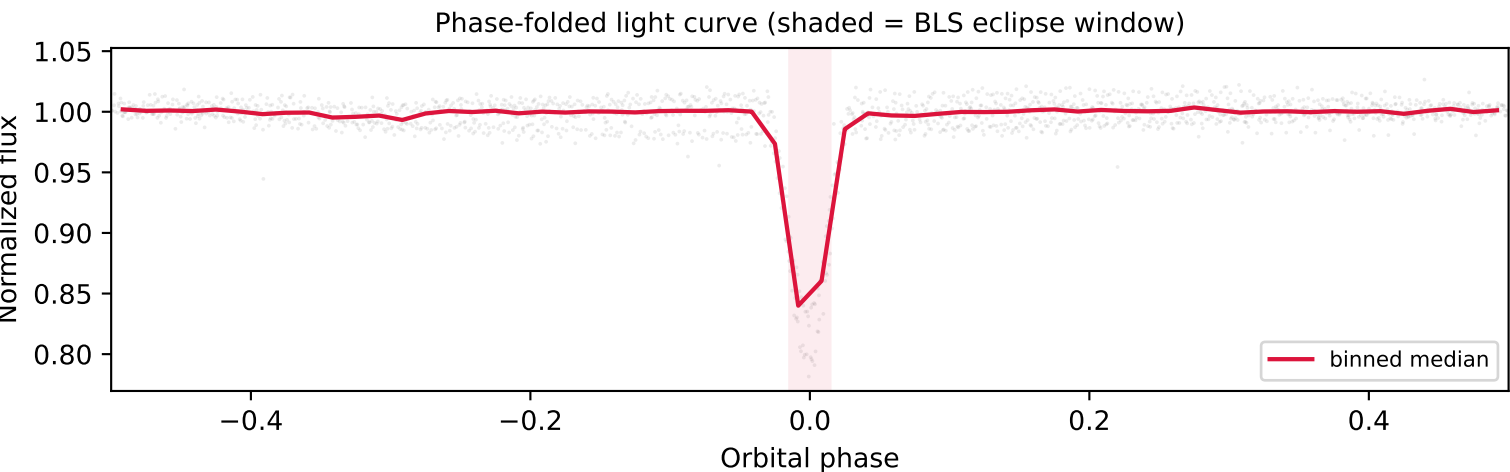


TIC 229146166

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

|                                                       |                                                         |                               |                     |
|-------------------------------------------------------|---------------------------------------------------------|-------------------------------|---------------------|
| BLS EPHEMERIS                                         |                                                         | CATALOG CROSS-MATCH (novelty) |                     |
| Period                                                | 1.995129 d (ALIASED, see below)                         | VSX                           | no match            |
| -> corrected period:                                  | 3.990257 d                                              | ASAS-SN                       | no match            |
| Epoch (t0)                                            | 1354.6820 BTJD                                          | Gaia eclipsing-binary table   | no match            |
| Duration                                              | 0.0580 d                                                | Gaia variable classification  | no match            |
| Depth                                                 | 15.426 % (154,264 ppm)                                  | TESS EB catalog (Prsa+2022)   | no match            |
| SNR                                                   | 446.3                                                   | ExoFOP TOI list               | no match            |
| INDEPENDENT CHECKS (f3i gates)                        |                                                         | Verdict: NOVEL                |                     |
| Eclipses observed                                     | 13 (gate >= 3)                                          |                               |                     |
| Robust depth ratio                                    | 1.03 (gate >= 0.50)                                     |                               |                     |
| Robust depth sigma                                    | 103.1 (gate >= 7.0)                                     |                               |                     |
| Odd/even z-score                                      | 10.40 (ALIAS FLAGGED)                                   |                               |                     |
| Secondary eclipse                                     | not detected                                            |                               |                     |
| PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging) |                                                         |                               |                     |
| Sector 2                                              | light-curve dip in this sector: 81.2 sigma              | t0 refined by 0.0000 phase    |                     |
| Difference-image peak SNR                             | 46.20 (gate >= 5.0)                                     | Compactness                   | 0.76 (gate >= 0.50) |
| Dimming centroid offset                               | 0.16 px (3 arcsec) (gate <= 1.0 px, i.e. on the target) |                               |                     |
| Nearest bright-enough neighbour in cutout:            | none within the cutout                                  |                               |                     |



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 10.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 3.990257 d, in any write-up.